

Math Module 1

1

The equation $d = 29w + 100$ gives the total amount of money d , in dollars, that James plans to save w weeks after he starts to save. What is the total amount of money, in dollars, he plans to save in 10 weeks?

- A) 340
- B) 390
- C) 440
- D) 490

2

Carrie has c dollars and Marjani has m dollars. Carrie has 3 times as many dollars as Marjani, and together they have a total of \$28.00. Which system of equations represents this situation?

- A) $c=3m$; $c+m=28$
- B) $m=3c$; $c+m=28$
- C) $c=28m$; $c+m=3$
- D) $m=28c$; $c+m=3$

3

$$x^2 = (29)(29)$$

What is the positive solution to the given equation?

- A) $\sqrt{29}$
- B) 29
- C) $2\sqrt{29}$
- D) $2 \cdot 29$

4

A car travels at a speed of at least 40 miles per hour but no more than 65 miles per hour for a certain part of a trip. Which inequality represents this situation, where x is the speed of the car, in miles per hour, on this part of the trip?

- A) $x \geq 40$
- B) $x \geq 65$
- C) $40 \leq x \leq 65$
- D) $x \leq 105$

5

$$x = 9$$

$$y = x^2 + 8$$

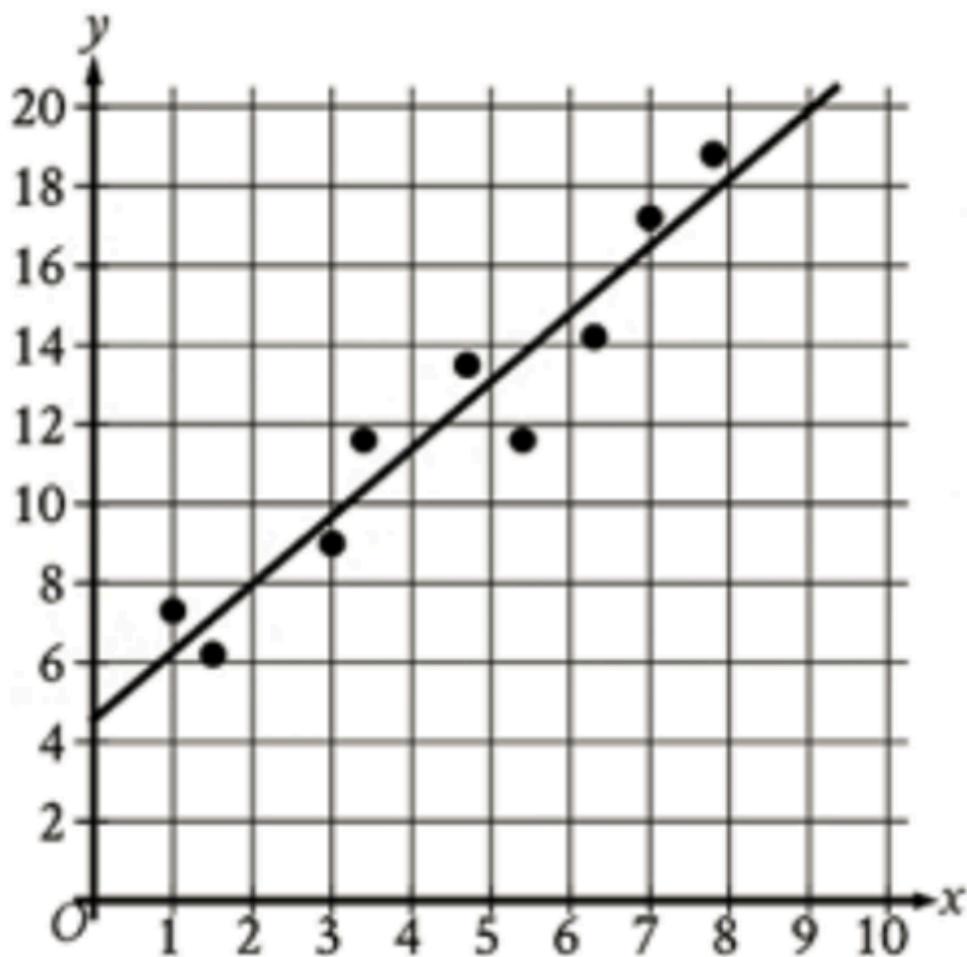
The graphs of the equations in the given system of equations intersect at the point (x, y) in the xy -plane. What is the value of y ?

- A) 9
- B) 26
- C) 81
- D) 89

6

The function f is defined by $f(x) = x^2 + x + 44$.
What is the value of $f(2)$?

7



The scatterplot shows the relationship between two variables, x and y . A line of best fit for the data is also shown. For how many of the 9 data points is the actual y -value less than the y -value predicted by the line of best fit?

- A) 2
- B) 4
- C) 5
- D) 7

8

Jasmin grows bean pods in her garden. This year, she harvested 490 bean pods and saved 10% of them to plant next year. How many of the harvested bean pods did Jasmin save to plant next year?

- A) 39
- B) 49
- C) 57
- D) 59

9

$6(3x - 4) = 42$

Which equation has the same solution as the given equation?

- A) $3x - 4 = 36$
- B) $3x - 4 = 7$
- C) $3x - 24 = 42$
- D) $3x - 24 = 48$

10

What is an x-intercept of the graph of the equation $y = (x + 8)(x - 11)(x + 12)$ in the xy-plane?

- A) (1,0)
- B) (0,8)
- C) (11,0)
- D) (0,12)

11

Each side of equilateral triangle S is multiplied by a scale factor of k to create equilateral triangle T. The length of each side of triangle T is greater than the length of each side of triangle S. Which of the following could be the value of k?

- A) 0
- B) $\frac{43}{44}$
- C) 1
- D) $\frac{44}{43}$

12

Which expression is equivalent to $2\left(x - \frac{7}{2}\right)(x + 6)$

- A) $x^2 + 5x - 21$
- B) $2x^2 - x - 21$
- C) $2x^2 + 5x - 42$
- D) $2x^2 - 42$

13

$$y = 9x - 15$$

$$5x + y = 41$$

The solution to the given system of equations is (x, y). What is the value of x + y?

14

A salesperson earns 2 points for each call made to a potential customer and 6 points for each visit made to a potential customer. Last week, the salesperson earned a total of 366 points. The equation $2x + 6y = 366$ represents this situation. Which is the best interpretation of 6y in this context?

- A) The number of calls the salesperson made to potential customers last week
- B) The number of visits the salesperson made to potential customers last week
- C) The total number of points the salesperson earned from calls made to potential customers last week
- D) The total number of points the salesperson earned from visits made to potential customers last week

15. One of the two equations in a system of linear equations is given. The system has infinitely many solutions. If the second equation in the system is $y = mx + b$, where m and b are constants, what is the value of b ?

A) -7

B) $-1/7$

C) $1/7$

D) 7

$$y = \frac{2}{5}x + 7$$

16

The kinetic energy, in joules, of an object with mass 75 kilograms traveling at a speed of v meters per second is given by the function K ,

$$K(v) = \frac{1}{2}(75)(v^2)$$

Which of the following is the best interpretation of $K(0) = 0$ in this context?

A) The kinetic energy of the object is 0 joules when the mass of the object is 0 kilograms.

B) The speed of the object is 0 meters per second when the mass of the object is 0 kilograms.

C) The kinetic energy of the object is 0 joules when the object is traveling at a speed of 0 meters per second.

D) The height of the object is 0 meters when the object is traveling at a speed of 0 meters per second.

17

In triangle ABC, angle B is a right angle. The length of side AB is $10\sqrt{69}$ and the length of side BC is $24\sqrt{69}$. What is the length of side AC?

A) $14\sqrt{69}$

B) $26\sqrt{69}$

C) $34\sqrt{69}$

D) $\sqrt{34 \cdot 69}$

18

x	y
-4	$n + 142$
-2	$n + 71$
0	n

A. $-\frac{71}{2}$

B. $-\frac{2}{71}$

C. $\frac{n+71}{-2}$

D. $\frac{2n-71}{2}$

The table shows three values of x and their corresponding values of y , where n is a constant, for the linear relationship between x and y . What is the slope of the line that represents this relationship in the xy -plane?

19

Square A has side lengths that are 192 times the side lengths of square B. The area of square A is k times the area of square B. What is the value of k ?

20

What is the value of $\tan\left(\frac{50\pi}{3}\right)$

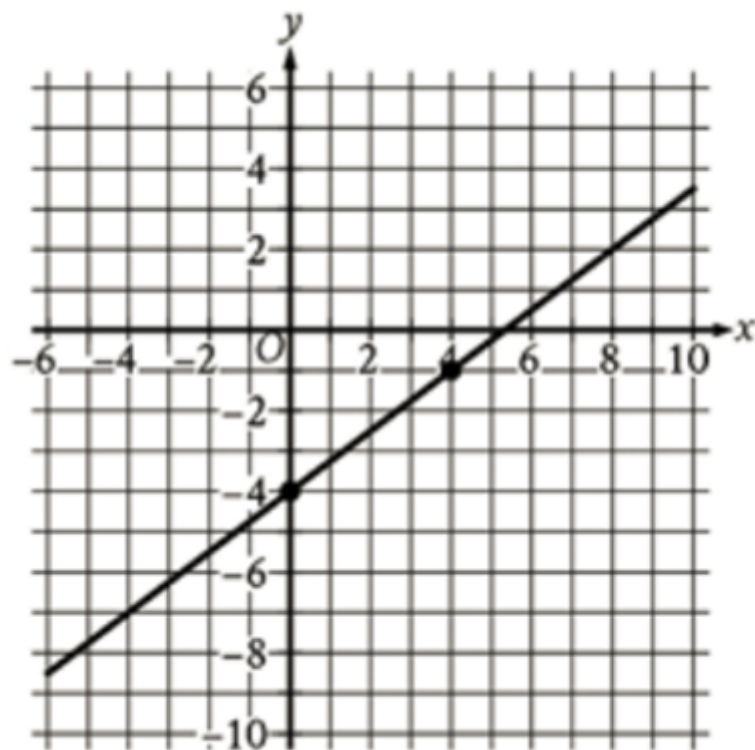
- A) $-\sqrt{3}$
- B) $-\sqrt{3}/3$
- C) $\sqrt{3}/3$
- D) $\sqrt{3}$

21

A conservation specialist hung artificial nesting structures each in the shape of a right rectangular prism for a species of native duck. Each structure has a height of 12 inches. The length of each structure's base is x inches, which is 1 inch more than the width of the structure's base. Which function V gives the volume of each structure, in cubic inches, in terms of the length of the structure's base?

- A) $V(x) = x(x + 12)(x + 1)$
- B) $V(x) = x(x + 12)(x - 1)$
- C) $V(x) = 12x(x + 1)$
- D) $V(x) = 12x(x - 1)$

22



Line r is shown in the xy -plane. Line t (not shown) is perpendicular to line r and passes through the point $(5, 8)$. What is the y -coordinate of the y -intercept of line t ?